

ENERGY STORAGE SOLUTION Enters Iron Age

In this series we shall be just scratching the surface of the iron-air battery saga. Their promise sparks a crucial question: Can this unassuming metal redefine our energy landscape?



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Lithium-ion batteries are utilised every day for torchlights to electric vehicles. However, global electric power generation is measured in gigawatts. Fossil fuel power stations are being phased out in favour of cheap, clean, and renewable energy, but they are very vulnerable to the vagaries of nature. A key roadblock for renewable energy to be viable is a long-term and reliable grid energy storage system, which cannot be adequately fulfilled by the current battery technology. Next-generation iron-air battery technology could revolutionise energy storage, unlocking tens of gigawatts of demand at any time from multiday grid storage systems using iron-air batteries.

Today, lithium-ion batteries are found in consumer products for energy storage. However, what works in a laptop is not the best solution for power grids, where energy output is measured in megawatts. Lithium-ion batteries perform well when consuming energy over a short period.

Researchers believe that this problem can be solved by employing iron-air batteries. Each battery, approximately the size of a washing machine, consists of about 50 iron-air cells. These cells breathe in air, transforming the iron into rust (iron oxide) and producing energy. During charging, the current reverses the oxidation, reducing iron oxide back into iron.

The primary task of these batteries is to bridge the gap when utilities need more power

during peak periods. They can store excess energy on sunny days to supplement supply on cloudy days. While lithium-ion batteries provide four hours of storage, iron-air batteries can deliver up to four days to bridge the gap, and they are cost-effective.

History

Metal-air batteries were first designed in 1878. This technology uses atmospheric oxygen as a cathode (electron receiver) and a cheap metal anode (electron giver) made of iron, aluminium, or zinc. In 1932, zinc-air batteries were the first type of metal-air battery widely used in hearing aids.

Three decades later, NASA developed iron-air batteries and created batteries with a 50-20Ah charge capacity capable of achieving less than 200 cycles with an energy density of 132-154Wh/kg.

They also reported that batteries involving air were more problematic due to difficulty in recharging. Oxygen reacts with the metal, creating a chemical that triggers the electrolysis process of discharging energy. Due to constant airflow, the battery would corrode quickly even when left unused, resulting in a very short shelf life.

The main advantage is that no metal dendrites are formed during charging. On the other hand, hydrogen is formed in substantial quantities.

The technique was utilised by the Swedish National Development Electric Corporation in 1977 to produce a 30kWh battery for traction applications with an energy density of 80Wh/kg at the 5-hour rate. Subsequently, in 1980, Westinghouse Electric Corporation used modified, improved iron sintered electrodes which could be made in modules

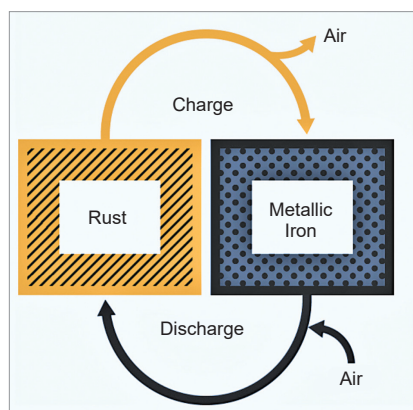


Fig. 1: Basic principle of iron-air battery